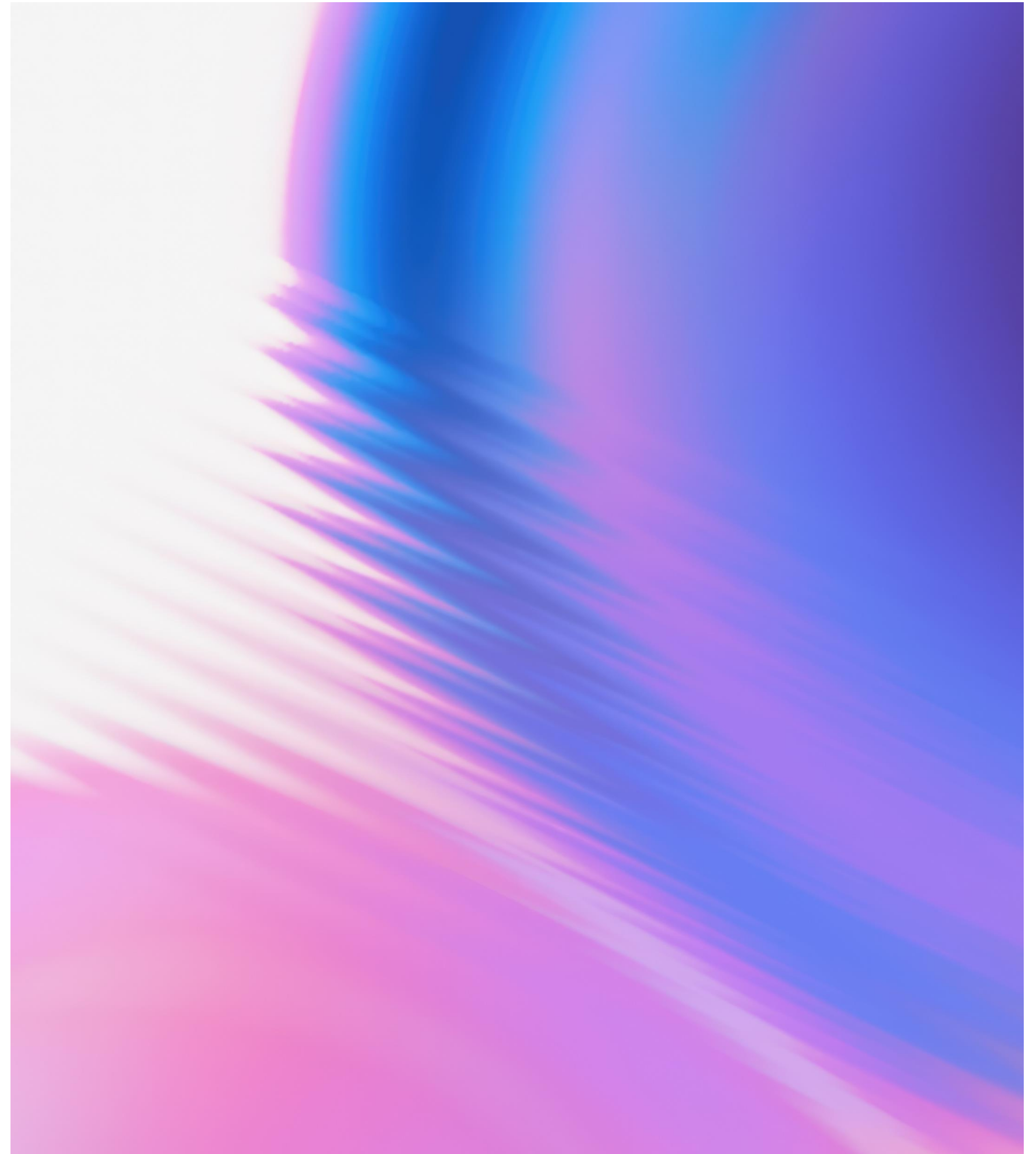
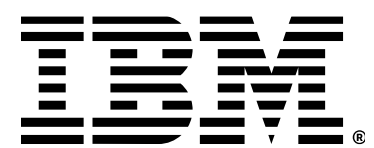


Quantum mechanics for computation

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Overview

- Classical information
- Quantum information
- Hands-on coding with Qiskit

Check out the "Basics of quantum information" course with John Watrous

<https://quantum.cloud.ibm.com/learning/en/courses/basics-of-quantum-information>

Classical states

A classical system X can be in one of a finite number of classical states. We use Σ to denote the set of classical states.

- If X is a bit, then $\Sigma = \{0, 1\}$
- If X is a six-sided die, then $\Sigma = \{\square, \square, \square, \square, \square, \square\}$

The state of X can also be probabilistic, representing uncertain knowledge. In general, the state can be described by a vector whose entries sum to 1.

Example:

If X is a bit and

$$\Pr(X = 0) = \frac{3}{4} \quad \text{and} \quad \Pr(X = 1) = \frac{1}{4}$$

this can be represented using the vector

$$\begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \end{pmatrix}$$

Dirac notation

Dirac notation is a notation for representing vectors and matrices widely used in quantum mechanics.

Dirac notation is not specific to quantum information, and can be used to represent classical information as well.

kets are column vectors

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

bras are row vectors

$$\langle 0| = (1 \quad 0) \quad \text{and} \quad \langle 1| = (0 \quad 1)$$

Dirac notation

A ket-bra gives a matrix

$$|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} (0 \quad 1) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

A bra-ket gives a scalar (number)

$$\langle a|b\rangle = \begin{cases} 1 & a = b \\ 0 & a \neq b. \end{cases}$$

kets are column vectors

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

bras are row vectors

$$\langle 0| = (1 \quad 0) \quad \text{and} \quad \langle 1| = (0 \quad 1)$$

Measuring probabilistic states

Measuring a system means observing it and recognizing what classical state it is in.

After we observe a probabilistic state, we gain knowledge about it, which we can use to update the probabilities we associate with each classical state.

Example:

If we flip a coin and don't look at the result, we can represent the state of the coin as

$$\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{1}{2} |\text{heads}\rangle + \frac{1}{2} |\text{tails}\rangle$$

If we look at it and see tails, then the state of the coin becomes

$$|\text{tails}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Classical operations

A deterministic operation transforms a classical state a into $f(a)$ for some function $f: \Sigma \rightarrow \Sigma$

When $\Sigma = \{0, 1\}$, there are four possible functions:

a	$f_1(a)$	a	$f_2(a)$	a	$f_3(a)$	a	$f_4(a)$
0	0	0	0	0	1	0	1
1	0	1	1	1	0	1	1

Any function can be written as a matrix M such that

$$M|a\rangle = |f(a)\rangle$$

Classical operations

a	$f_1(a)$
0	0
1	0

a	$f_2(a)$
0	0
1	1

a	$f_3(a)$
0	1
1	0

a	$f_4(a)$
0	1
1	1

$$M|a\rangle = |f(a)\rangle$$

$$M_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_4 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

$$M = \sum_{a \in \Sigma} |f(a)\rangle \langle a|$$

Classical operations

In general, a classical operation M doesn't need to be deterministic. The only requirement is that it maps probability vectors to probability vectors.

Such operations are described by stochastic matrices.

A square matrix is *stochastic* if its entries are nonnegative and each column sums to one. In other words, each column is a probability vector.

A probabilistic operation can be interpreted as a random choice among deterministic operations.

Example:

$$\begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

Multiple classical systems

If there are two systems:

- X with state set Σ
- Y with state set Γ

then the state set of the combined system (X, Y) is the Cartesian product

$$\Sigma \times \Gamma = \{(a, b) : a \in \Sigma \text{ and } b \in \Gamma\}$$

Example:

If X and Y are both bits, then $\Sigma = \Gamma = \{0, 1\}$ and

$$\Sigma \times \Gamma = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$$

We usually write these as bitstrings,

$$\Sigma \times \Gamma = \{00, 01, 10, 11\}$$

The tensor product

In the language of vectors and matrices, multiple systems are combined using the *tensor product*.

Given two vectors

$$|\phi\rangle = \sum_{a \in \Sigma} \alpha_a |a\rangle \quad \text{and} \quad |\psi\rangle = \sum_{b \in \Gamma} \beta_b |b\rangle$$

The tensor product $|\phi\rangle \otimes |\psi\rangle$ is defined as

$$|\phi\rangle \otimes |\psi\rangle = \sum_{(a,b) \in \Sigma \times \Gamma} \alpha_a \beta_b |ab\rangle$$

The tensor product

$$\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_m \end{pmatrix} \otimes \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} = \begin{pmatrix} \alpha_1 \beta_1 \\ \vdots \\ \alpha_1 \beta_k \\ \alpha_2 \beta_1 \\ \vdots \\ \alpha_2 \beta_k \\ \vdots \\ \alpha_m \beta_1 \\ \vdots \\ \alpha_m \beta_k \end{pmatrix}$$

Given two vectors

$$|\phi\rangle = \sum_{a \in \Sigma} \alpha_a |a\rangle \quad \text{and} \quad |\psi\rangle = \sum_{b \in \Gamma} \beta_b |b\rangle$$

The tensor product $|\phi\rangle \otimes |\psi\rangle$ is defined as

$$|\phi\rangle \otimes |\psi\rangle = \sum_{(a,b) \in \Sigma \times \Gamma} \alpha_a \beta_b |ab\rangle$$

The tensor product

It is common to omit the $\otimes$ symbol and just write $|\phi\rangle|\psi\rangle$ instead of $|\phi\rangle \otimes |\psi\rangle$.

The tensor product has the important property that it is *bilinear*.

Linearity in the first argument:

$$\begin{aligned}(|\phi_1\rangle + |\phi_2\rangle)|\psi\rangle &= |\phi_1\rangle|\psi\rangle + |\phi_2\rangle|\psi\rangle \\ (\alpha|\phi\rangle)|\psi\rangle &= \alpha(|\phi\rangle|\psi\rangle)\end{aligned}$$

Linearity in the second argument:

$$\begin{aligned}|\phi\rangle(|\psi_1\rangle + |\psi_2\rangle) &= |\phi\rangle|\psi_1\rangle + |\phi\rangle|\psi_2\rangle \\ |\phi\rangle(\alpha|\psi\rangle) &= \alpha(|\phi\rangle|\psi\rangle)\end{aligned}$$

The tensor product

The tensor product of matrices is defined similarly.

Given two matrices

$$M = \sum_{a,b \in \Sigma} \alpha_{ab} |a\rangle \langle b| \quad \text{and} \quad N = \sum_{c,d \in \Gamma} \beta_{cd} |c\rangle \langle d|$$

The tensor product $M \otimes N$ is defined as

$$M \otimes N = \sum_{a,b \in \Sigma} \sum_{c,d \in \Gamma} \alpha_{ab} \beta_{cd} |ac\rangle \langle bd|$$

The tensor product

$$\begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1m} \\ \vdots & \ddots & \vdots \\ \alpha_{m1} & \cdots & \alpha_{mm} \end{pmatrix} \otimes \begin{pmatrix} \beta_{11} & \cdots & \beta_{1k} \\ \vdots & \ddots & \vdots \\ \beta_{k1} & \cdots & \beta_{kk} \end{pmatrix}$$

$$= \begin{pmatrix} \alpha_{11}\beta_{11} & \cdots & \alpha_{11}\beta_{1k} & \cdots & \alpha_{1m}\beta_{11} & \cdots & \alpha_{1m}\beta_{1k} \\ \vdots & \ddots & \vdots & \cdots & \vdots & \ddots & \vdots \\ \alpha_{11}\beta_{k1} & \cdots & \alpha_{11}\beta_{kk} & & \alpha_{1m}\beta_{k1} & \cdots & \alpha_{1m}\beta_{kk} \\ & \vdots & & \ddots & & \vdots & \\ \alpha_{m1}\beta_{11} & \cdots & \alpha_{m1}\beta_{1k} & & \alpha_{mm}\beta_{11} & \cdots & \alpha_{mm}\beta_{1k} \\ \vdots & \ddots & \vdots & \cdots & \vdots & \ddots & \vdots \\ \alpha_{m1}\beta_{k1} & \cdots & \alpha_{m1}\beta_{kk} & & \alpha_{mm}\beta_{k1} & \cdots & \alpha_{mm}\beta_{kk} \end{pmatrix}$$

Representation of computer programs

The state space of N bits is the set of all bitstrings of length N , which has size 2^N (exponential in N).

For 10 bits:

```
0000000000
0000000001
0000000010
0000000011
      ⋮
1111111110
1111111111
```

The probability vector for a system of 500 bits has length 2^{500} , which is more than the number of atoms in the observable universe.

Computer programs don't explicitly store all of the probabilities.

Representation of computer programs

```
import random

n_bits = 500
bits = random.getrandbits(n_bits)
print(format(bits, f"0{n_bits}b"))
```

```
11100100000000001001000001100101110001100101100001
10101101001110110001001111000111100111001010001101
00000001011000001011011100111001001110010001100101
10100100010000000110000011011101100011010111010011
01110110001101010010110111101010000001110000100101
10111111110001001100110100100000110000111010001111
01101001001000001110110001100110001111111011111100
00100011011101111001011000110111000000010100010111
00010110000111111010111110000011111000010000110001
00011110011111110000100000100100100100110011000001
```

This Python script samples a bitstring from the probability distribution

$$\left\{ \begin{pmatrix} \frac{1}{2^{500}} \\ \frac{1}{2^{500}} \\ \vdots \\ \frac{1}{2^{500}} \end{pmatrix} \right\} 2^{500} \text{ entries}$$

Quantum states

A quantum system also has a classical state set.
A *qubit* has the same state set as a bit, $\Sigma = \{0, 1\}$.

The state of a quantum system is described by a vector of complex numbers. The sum of the absolute values squared must be 1.

$$\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{pmatrix} \text{ such that } \alpha \in \mathbb{C}, \sum_k |\alpha_k|^2 = 1$$

Examples of qubit states:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

$$\begin{pmatrix} \frac{1+2i}{3} \\ -\frac{2}{3} \end{pmatrix} = \frac{1+2i}{3} |0\rangle - \frac{2}{3} |1\rangle$$

Quantum states

The *Euclidean norm* of a vector

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{pmatrix}$$

is defined as

$$\| |\psi\rangle \| = \sqrt{\sum_{k=1}^N |\alpha_k|^2}$$

So a quantum state vector must satisfy

$$\| |\psi\rangle \| = 1$$

In other words, it's a unit vector under the Euclidean norm. Equivalently,

$$\begin{pmatrix} |\alpha_1|^2 \\ \vdots \\ |\alpha_N|^2 \end{pmatrix}$$

is a probability vector.

Measuring quantum states

Observing a quantum state reveals a classical state.

The probability of the state is equal to the absolute value squared of the corresponding state vector entry.

After measurement, the state of the system changes to the basis vector corresponding to the classical outcome.

For example, upon measuring 0, the quantum state becomes $|0\rangle$.

Example:

If we measure the state

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

Then

$$\text{Pr}(\text{outcome is } 0) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

$$\text{Pr}(\text{outcome is } 1) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

Quantum operations

A quantum operation U is represented by a matrix, and it must map valid state vectors to other valid state vectors.

More generally, valid operations preserve the Euclidean norm:

$$\|U|\psi\rangle\| = \||\psi\rangle\|$$

for any vector $|\psi\rangle$.

Such matrices are called *unitary* matrices.

Unitary matrices satisfy

$$U^\dagger U = I$$

Examples of single-qubit operations

Pauli matrices:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Hadamard gate:

$$H = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

Phase gate:

$$P_{\theta} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$

Multiple quantum systems

As with classical systems, multiple quantum systems are combined using the tensor product.

Example:

The vector space of two qubits is spanned by four states: $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$

An example of a two-qubit state is

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Comparison of classical and quantum information

	Classical	Quantum
State representation	Probability vector $\begin{pmatrix} p_1 \\ \vdots \\ p_N \end{pmatrix}$ such that $0 \leq p_k \leq 1,$ $\sum_k p_k = 1$	State vector $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{pmatrix}$ such that $\alpha \in \mathbb{C},$ $\sum_k \alpha_k ^2 = 1$
Operations	Stochastic matrices	Unitary matrices
Multiple systems	Tensor product	Tensor product
Efficient description	Python script or equivalent	Quantum circuit

